

TARGET-90

JKBOSE HUMANITIES CLASS- 11TH

GRAVITATION

One Shot

PHYSICS

By-Wajahid Sir (Gabbar Sir)





Topics to be Covered

A Universal Law of Gravitation

B Vector Form of Gravitational Force

C Gravitational Field

D Variation of g

E Gravitational Potential Energy



Topics to be Covered

F Kepler Law of Planetary Motion

G Escape Velocity

H Orbital Velocity ^{of} Satellite

I ...

J ...

GRAVITATION

Kepler's laws of planetary motion universal law of gravitation. Acceleration due to gravity and its variation with altitude and depth. Gravitational potential energy and gravitational potential, escape speed, orbital velocity of a satellite.

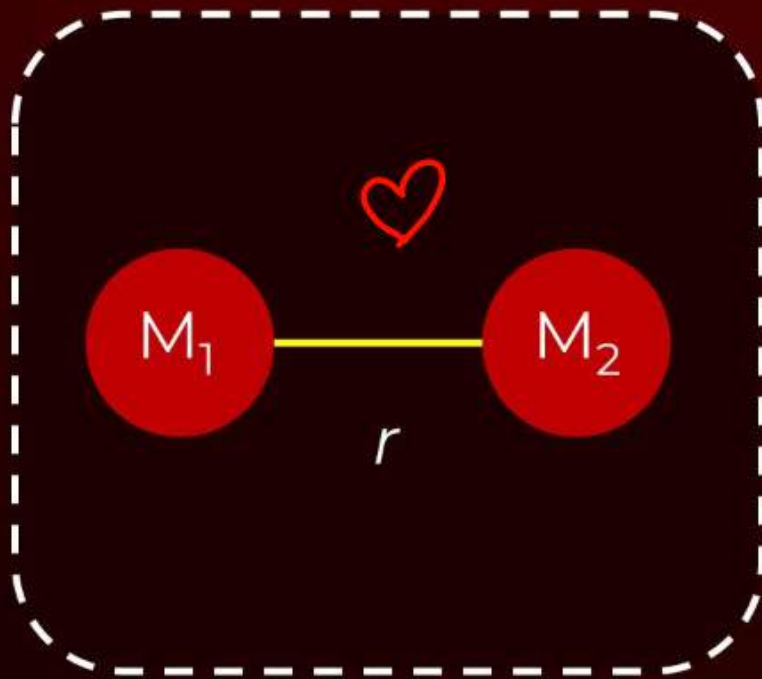




Universal Law Of Gravitation

- ✦ Every mass attracts every other mass with the Force which is directly proportional to product of mass.

→ of attraction



$$F \propto m_1 \times m_2 \text{ — ①}$$

$$F \propto \frac{1}{r^2} \text{ — ②}$$

$$F \propto \frac{m_1 m_2}{r^2}$$

and inversely
Proportion to
Square of
distance b/w
them

$$F = G \frac{m_1 m_2}{r^2}$$

G = Gravitation constant





Units and dimensions

★ Units (G)

$$F = G \frac{m_1 m_2}{r^2}$$

$$N = G \frac{\text{kg kg}}{\text{m}^2}$$

$$\text{Nm}^2 = G \text{ kg}^2$$

$$\frac{\text{Nm}^2}{\text{kg}^2} = G$$

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$



★ Dimensions (G)

H/W

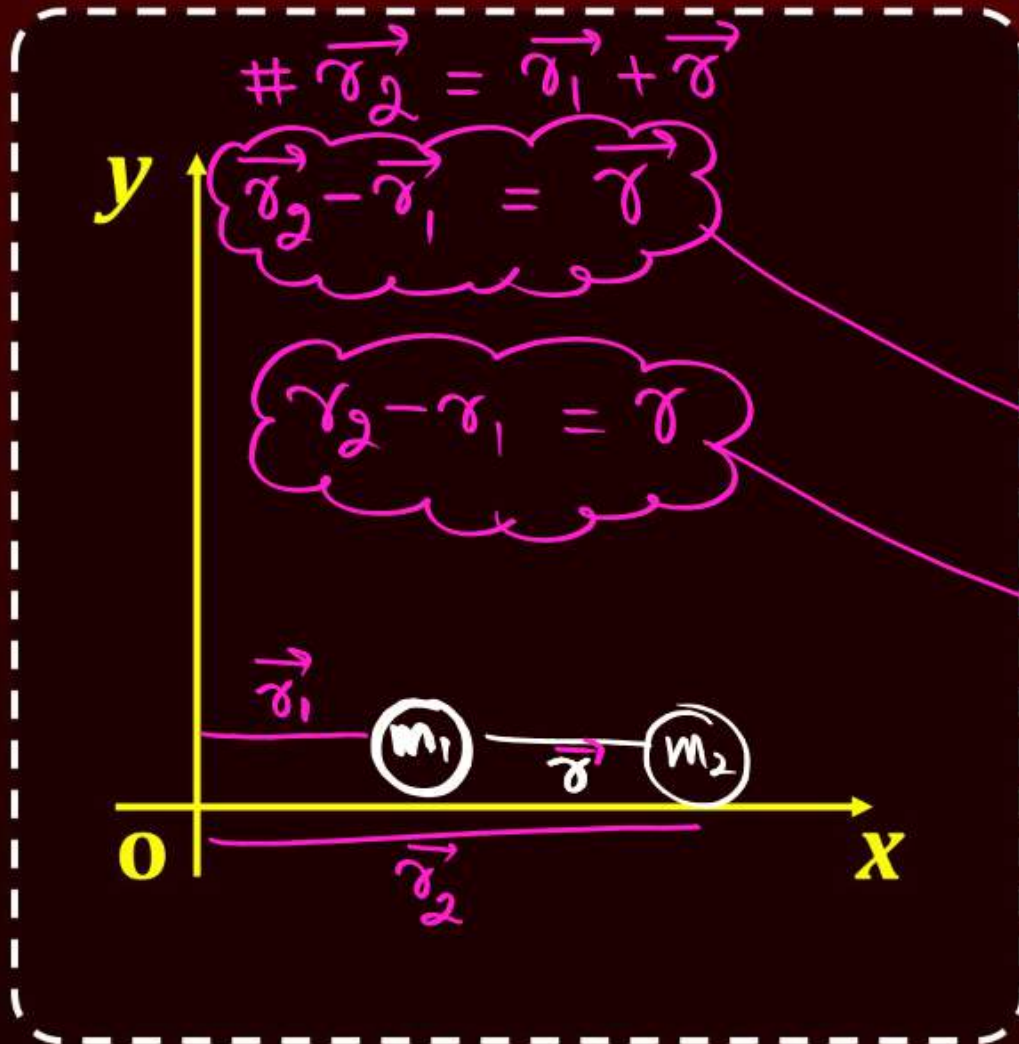


$$G = M^{-1} L^3 T^{-2}$$



Vector Form of Gravitational Force

- Newton tell us about the Force ($F = G \frac{m_1 m_2}{r^2}$)
- $F \Rightarrow$ vector
- $F = \text{magnitude} \times \text{direction}$



$$\vec{F} = G \frac{m_1 m_2}{r^2} \hat{n} \quad \hat{n} = \frac{\vec{r}}{r}$$

$$\vec{F} = G \frac{m_1 m_2}{r^2} \frac{\vec{r}}{r}$$

$$\vec{F} = G \frac{m_1 m_2}{r^3} \vec{r}$$

$$\vec{F} = \frac{G m_1 m_2}{(\vec{r}_2 - \vec{r}_1)^3} (\vec{r}_2 - \vec{r}_1)$$

Vector form of
Gravitation force.



Acceleration Due to Gravity/Gravitation Field

g

- Gravitational field is the area around the mass where force of attraction can be experienced.
- Gravitational field is the force acting per unit

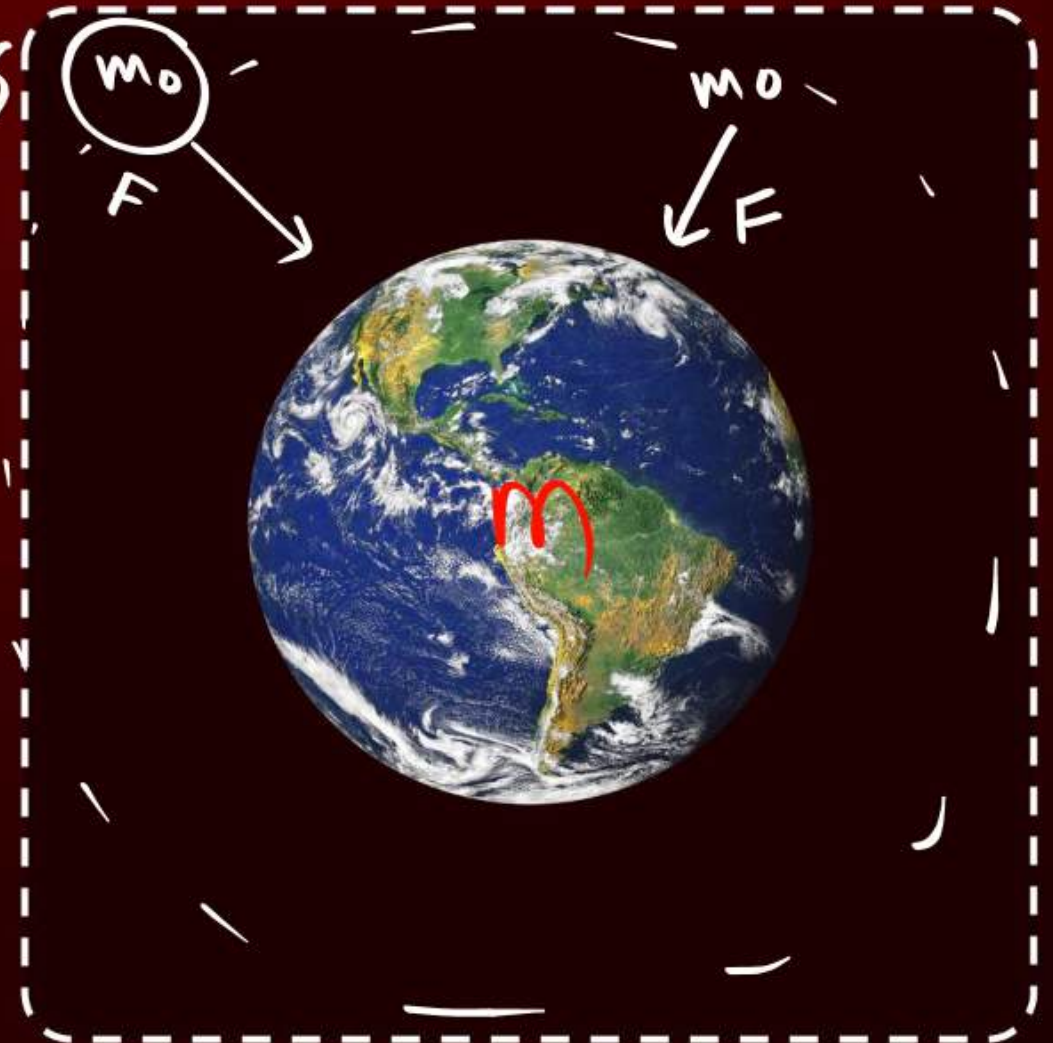
m_0

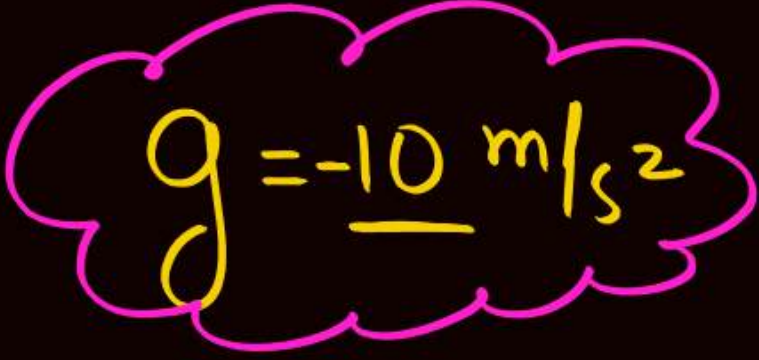
$$g = \frac{F}{m_0}$$

$$F = \frac{GMm_0}{r^2}$$

$$g = \frac{\frac{GMm_0}{r^2}}{m_0}$$

$$g = \frac{GM}{r^2}$$




$$g = \underline{-10 \text{ m/s}^2}$$

$$g = -9.8 \text{ m/s}^2$$



Acceleration Due to Gravity/Gravitation Field

due to earth
we know that the
mass of earth
($m = 6 \times 10^{24} \text{ kg}$) and
radius of earth = $6 \times 10^6 \text{ m}$

$$g_e = \frac{G m_e}{R_e^2}$$

$$g_e = \frac{6 \times 10^{-11} \times 6 \times 10^{24}}{(6 \times 10^6)^2}$$

$$g_e = \frac{\cancel{6}^2 \times 10^{-11} \times 10^{24}}{\cancel{6}^2 \times (10^6)^2}$$

$$g_e = \frac{10^{-11} \times 10^{24}}{10^{12}}$$

$$g_e = \frac{10^{-11+24}}{10^{12}}$$

$$g_e = \frac{10^{13} 10^1}{10^{12}}$$

$$g_e = 10 \text{ m/s}^2$$

$$g = -10 \text{ m/s}^2 \downarrow \text{Ⓜ}$$





Density Of Earth

$$V_s = \frac{4}{3} \pi R^3$$

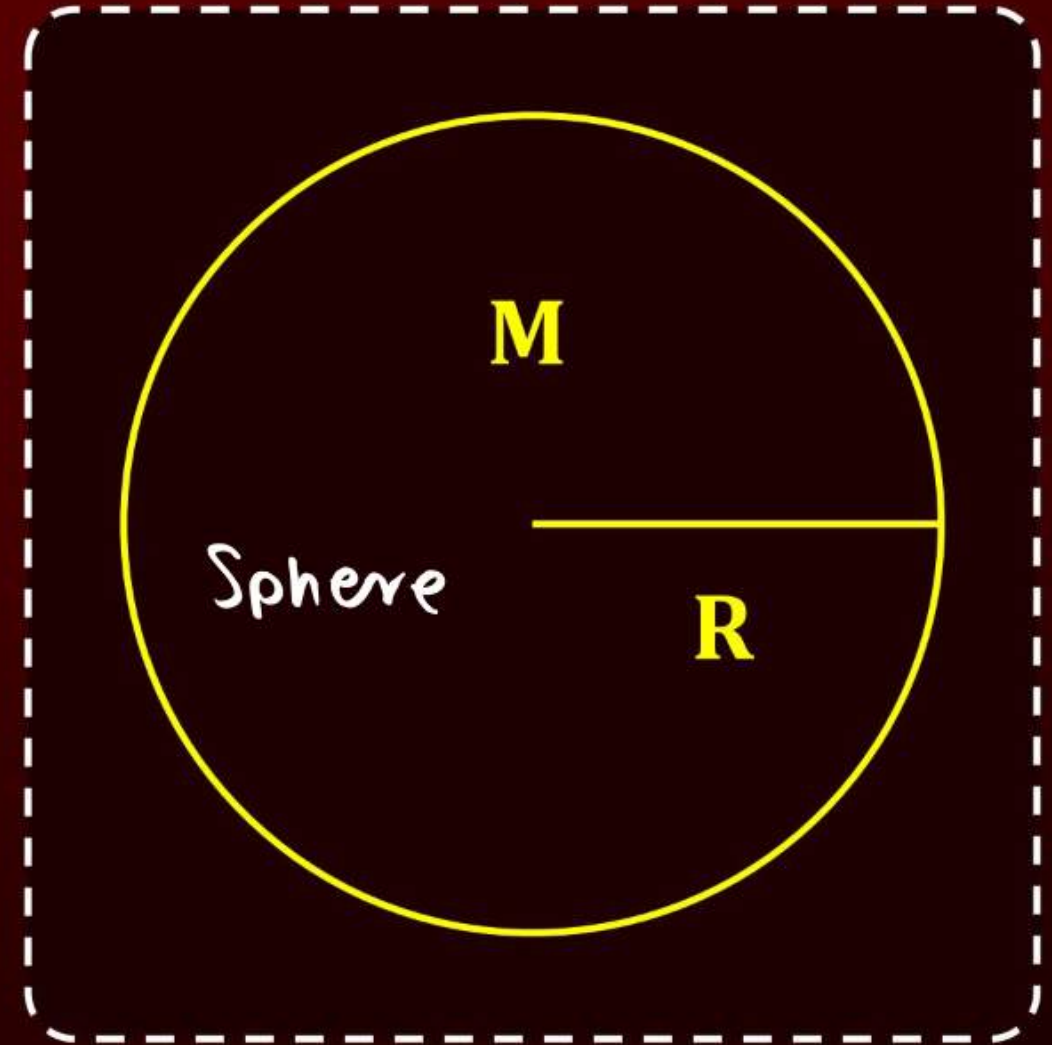
$$\rho = \frac{m}{v}$$

$$\rho = \frac{M}{\frac{4}{3} \pi R^3}$$

$$\rho = \frac{m}{\frac{4}{3} \pi R^3}$$

$$\frac{4}{3} \pi R^3 \rho = m$$

$$\rho = \frac{m}{\frac{4}{3} \pi R^3}$$



FORMULA OF (g) in term of density (ρ)

$$g = \frac{G(m)}{R^2} \quad \text{--- (I)} \quad m = \frac{4}{3} \pi R^3 \rho$$

$$g = \frac{G \frac{4}{3} \pi R^3 \rho}{R^2}$$

$$g = G \left(\frac{4}{3} \pi \right) R \rho$$

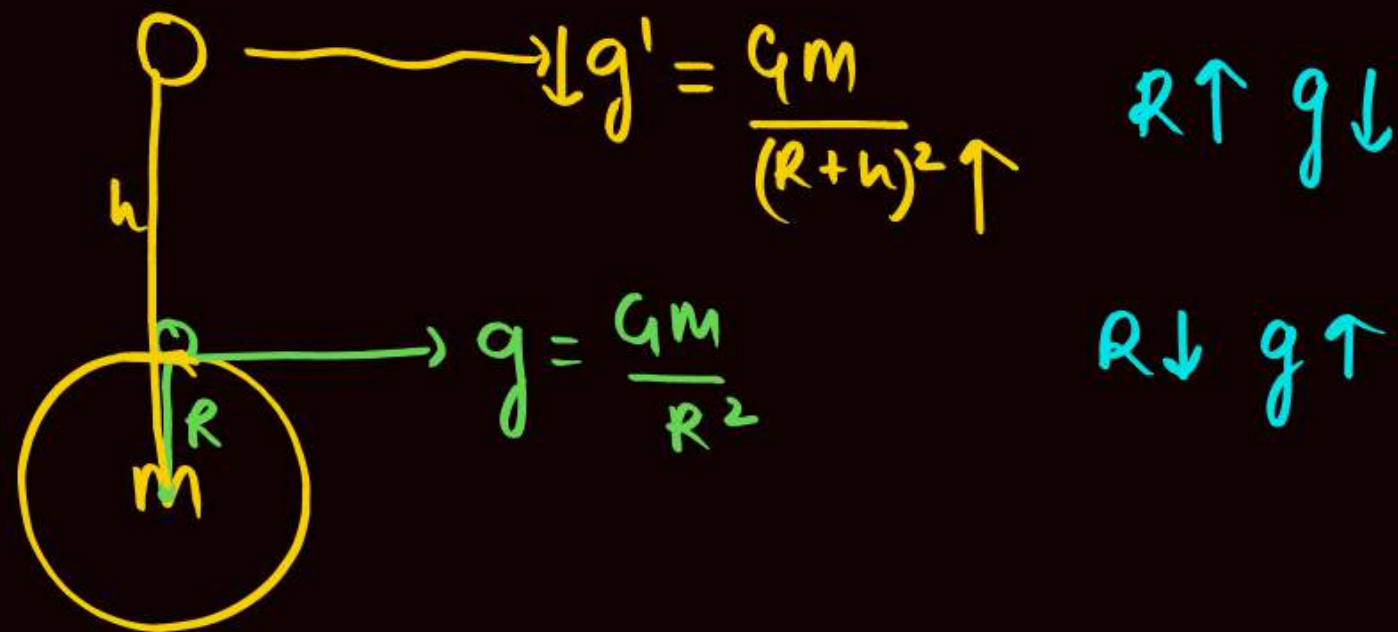
$$g = \frac{4}{3} \pi G \rho R \quad \text{--- (II)}$$

Density

Q. Is value of g is constant

Sol NO

$$\uparrow \downarrow g \uparrow \downarrow = \frac{Gm}{R^2} \uparrow \downarrow$$



Note :- the value of g is not constant



Variation of (g)

→ Change

let us consider the earth of mass (m) and Radius (R)

PYQ

2024

- ① With Height
on surface of earth

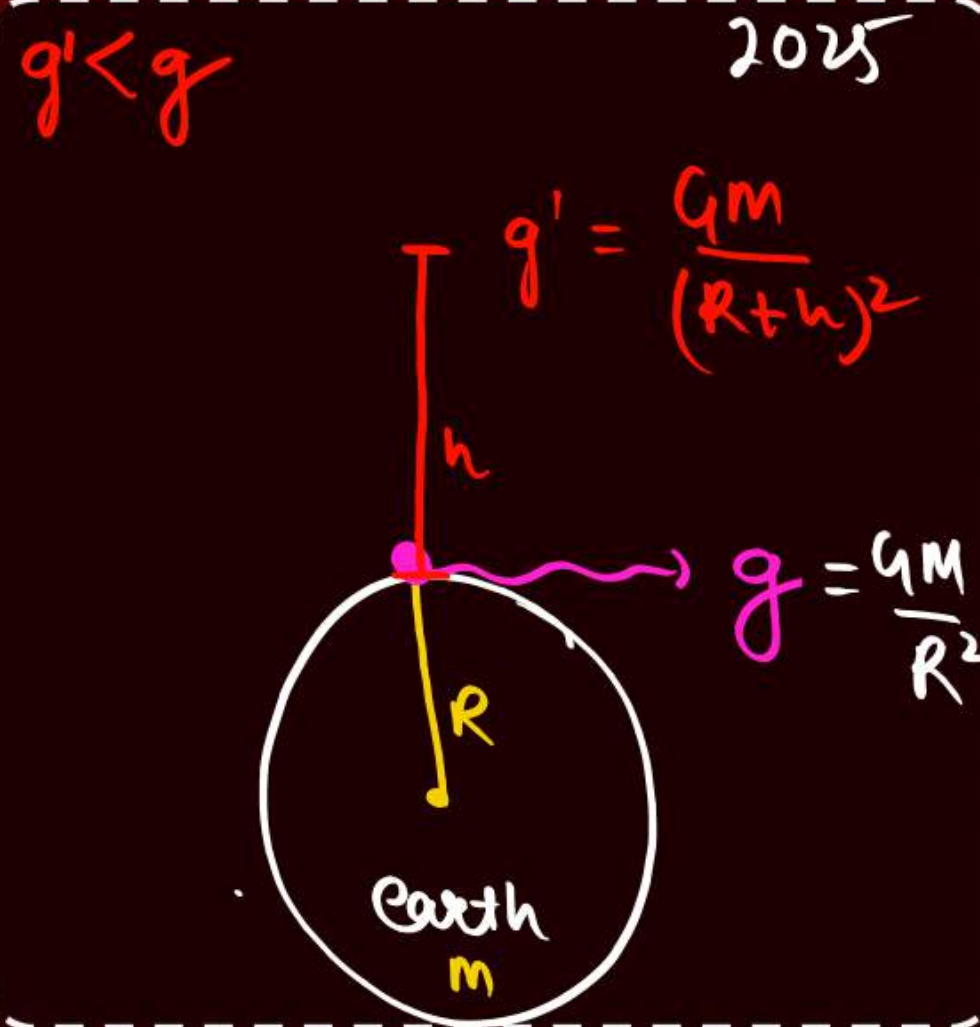
$$g = \frac{Gm}{R^2} \text{ --- (I)}$$

on some height

$$g' = \frac{Gm}{(R+h)^2} \text{ --- (II)}$$

Divide (II)
and (I)

$$\frac{g'}{g} = \frac{\frac{Gm}{(R+h)^2}}{\frac{Gm}{R^2}}$$



$$\frac{g'}{g} = \frac{\frac{Gm}{(R+h)^2}}{\frac{Gm}{R^2}} \quad \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c}$$

$$\frac{g'}{g} = \frac{\cancel{Gm}}{(R+h)^2} \times \frac{R^2}{\cancel{Gm}}$$

$$\frac{g'}{g} = \frac{R^2}{(R+h)^2}$$

$$\frac{g'}{g} = \frac{R^2}{\left(R + \frac{h}{R}\right)^2}$$

$$\frac{g'}{g} = \frac{R^2}{\left(\underline{R} + \frac{Rh}{\underline{R}}\right)^2}$$

$$\frac{g'}{g} = \frac{R^2}{\left(R\left(1 + \frac{h}{R}\right)\right)^2}$$

$$\frac{g'}{g} = \frac{\cancel{R^2}}{\cancel{R}^2 \left(1 + \frac{h}{R}\right)^2}$$

$$\frac{g'}{g} = \frac{1}{\left(1 + \frac{h}{R}\right)^2}$$

$$\frac{g'}{g} = \left(1 + \frac{h}{R}\right)^{-2}$$

$$\frac{g'}{g} = \left(1 + \frac{h}{R}\right)^{-2}$$

Binomial theorem

$$(1+x)^n = (1+nx)$$

$$\frac{g'}{g} = \left(1 + \frac{-2h}{R}\right)$$

$$\frac{g'}{g} = \left(1 - \frac{2h}{R}\right)$$

$$\frac{g'}{g} = \left(1 - \frac{2h}{R}\right)$$

$$g' = g \left(1 - \frac{2h}{R}\right)$$

$$g' = g \left(1 - \frac{2h}{R}\right)$$

$$g' = \underline{g} - g \frac{2h}{R}$$

$$g = 10$$

$$g' = 10 - 2$$

$$g' = 8$$

$$g' = 8$$

$$g > g'$$



II Variation of (g)

let us consider the earth of mass (m) and Radius (R)

With depth

$$\# R = d + x$$

$$(R - d) = x$$

II ✨ With Depth

- On the surface

$$g = \frac{4}{3} \pi G \rho R \quad \text{--- (I)}$$

- In Depth

$$g' = \frac{4}{3} \pi G \rho (R - d) \quad \text{--- (II)}$$

$$\downarrow g' = \frac{4}{3} \pi G \rho \underline{x} \downarrow$$

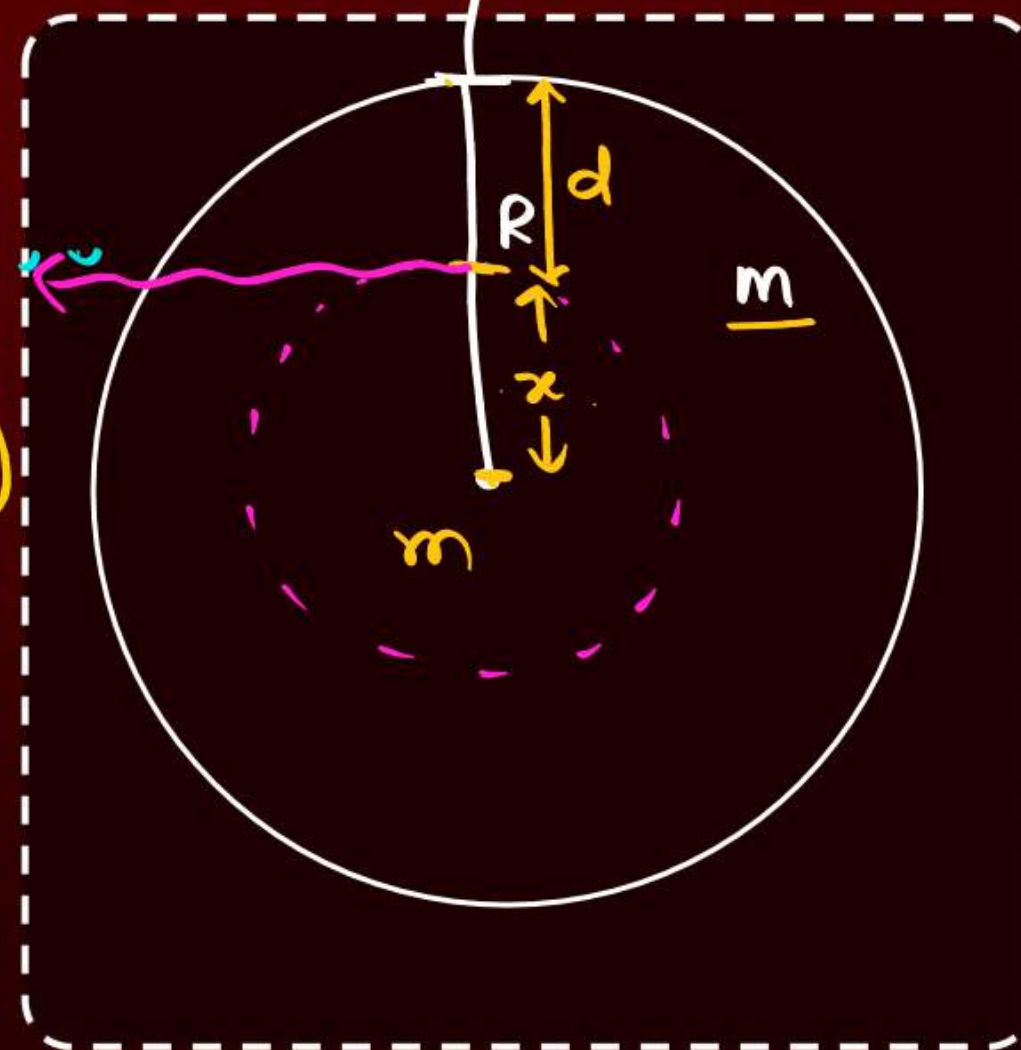
$$g' = \frac{4}{3} \pi G \rho (R - d)$$

PYQ
2023
2021
2026

$$\downarrow g = \frac{Gm}{R^2} \uparrow$$



$$\rightarrow g = \frac{4}{3} \pi G \rho R$$



Divide (II) and (I)

$$\frac{g'}{g} = \frac{\cancel{\frac{4}{3}} \cancel{\pi} \cancel{g} \cancel{r} (R-d)}{\cancel{\frac{4}{3}} \cancel{\pi} \cancel{g} \cancel{r} R}$$

$$\frac{g'}{g} = \frac{(R-d)}{R}$$

$$\frac{g'}{g} = \frac{\cancel{1} R}{\cancel{R}} - \frac{d}{R}$$

$$\frac{g'}{g} = 1 - \frac{d}{R}$$

$$g' = g \left(1 - \frac{d}{R} \right)$$

$$g \gg \gg g'$$

$$g' = g \left(1 - \frac{d}{R} \right)$$

$$\left(g' = g - \frac{g d}{R} \right) \quad g = 10$$

$$g' = 10 - 2$$

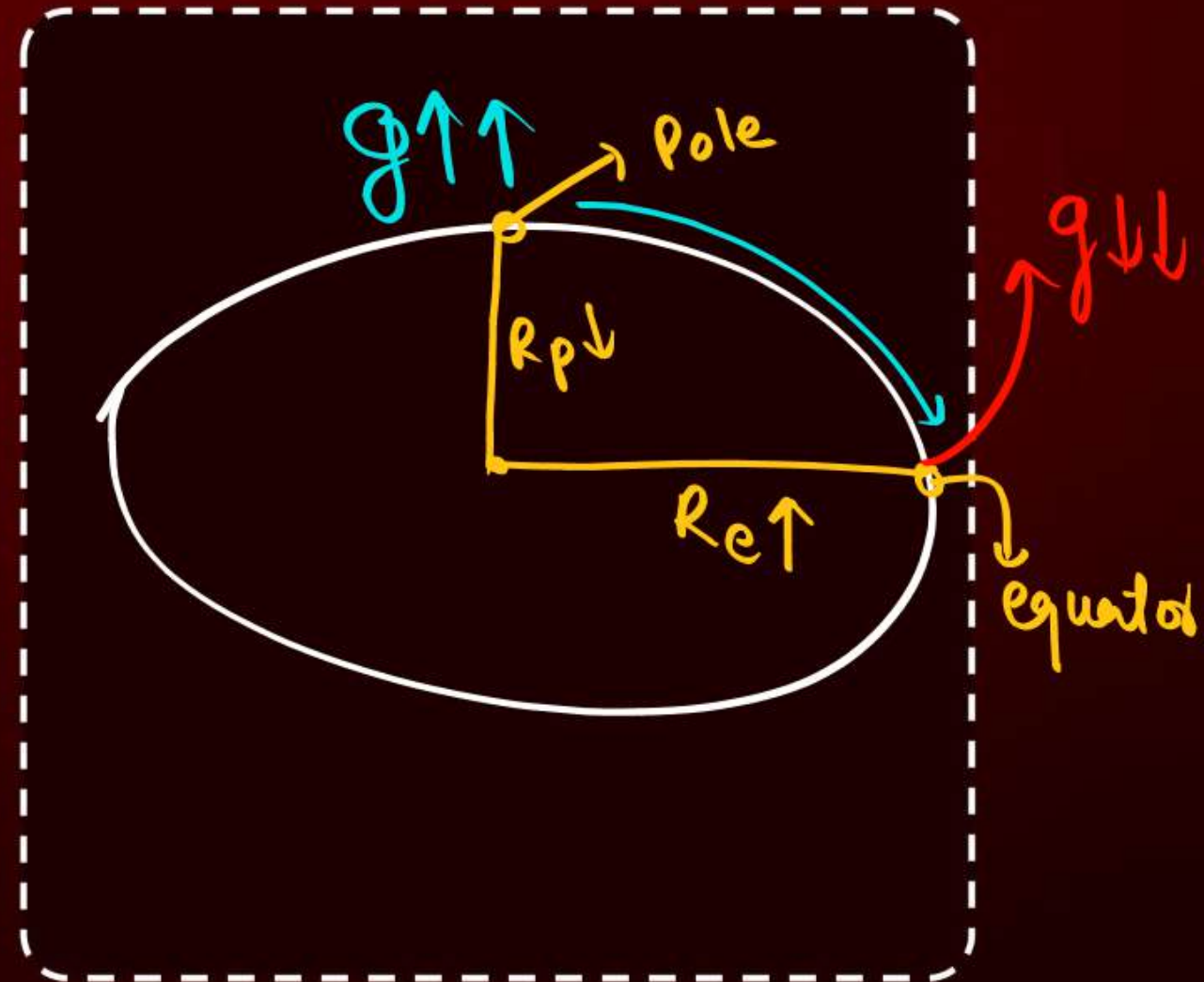
$$g' = 8$$



Variation of (g)

2025
mcq

- ③ Due to the Shape of Earth
the shape of earth is not perfect sphere
- # pole $\uparrow g_p = \frac{Gm}{(R_p)^2} \downarrow$
- # Equator $\downarrow g_e = \frac{Gm}{(R_e)^2} \uparrow$





Gravitational Potential Energy

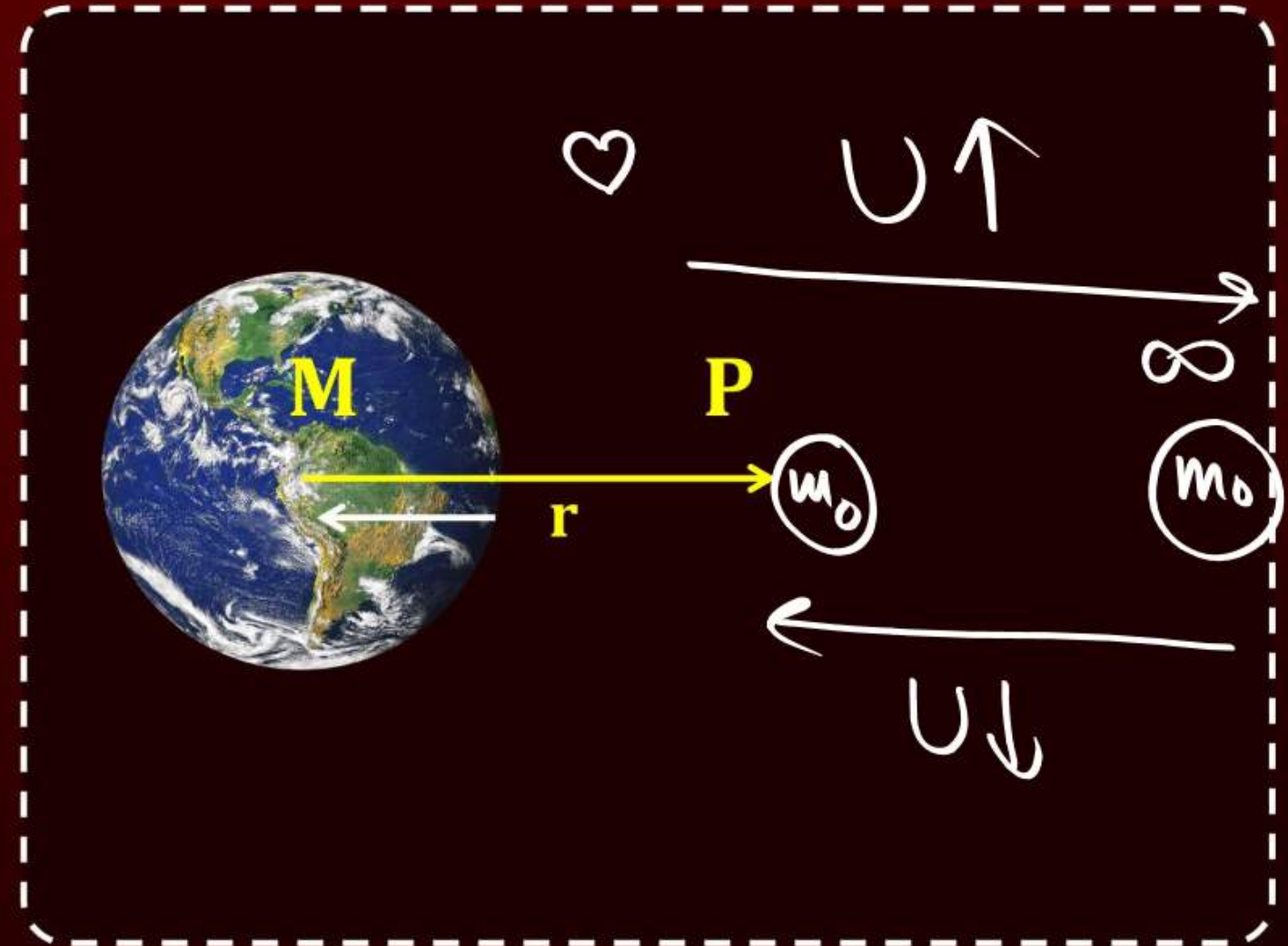
→ 2024
2023



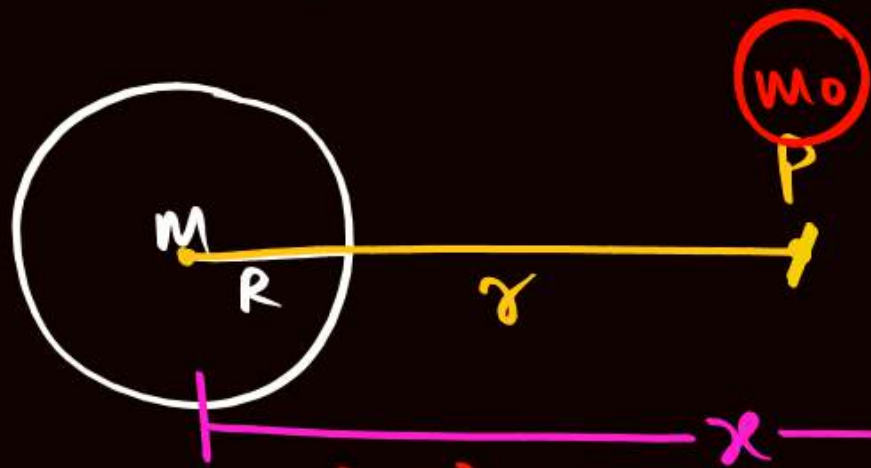
Energy due to Zabardasti

- ✦ Potential energy is due to the position of particle.
or
- ✦ The amount of work in bringing the unit mass from ∞ to that point (P)

$$U = -G \frac{M m_0}{r}$$



(1) let us consider Spherical Body of mass (m) and Radius (R)

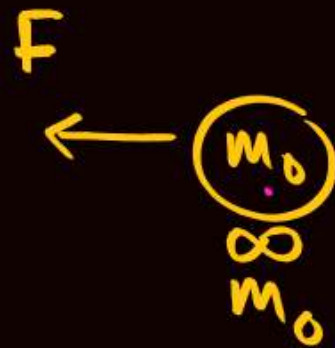


$W = \vec{F} \cdot \vec{x}$

$dw = F dx$
on integrating b/s

$$\int dw = \int_{\infty}^r F dx$$

$$\int dw = \int_{\infty}^r \underline{F} dx$$



$$W = \int_{\infty}^r \frac{GMm_0}{x^2} dx$$

$$W = GMm_0 \int_{\infty}^r \frac{1}{x^2} dx$$

$F = \frac{GMm_0}{x^2}$ (variable)
 $\downarrow$
 change
 $\downarrow$
 work
 $\downarrow$
 Integration

$$W = GMm_0 \int_{\infty}^r x^{-2} dx$$

$$W = G M m_0 \int_{\infty}^r x^{-2} dx \quad \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$W = G M m_0 \left[\frac{x^{-2+1}}{-2+1} \right]_{\infty}^r$$

$$W = G M m_0 \left[\frac{x^{-1}}{-1} \right]_{\infty}^r$$

$$W = \frac{G M m_0}{-1} \left[x^{-1} \right]_{\infty}^r$$

$$W = \frac{G M m_0}{-1} \left[\frac{1}{r} \right]_{\infty}^r$$

$$W = -G M m_0 \left[\frac{1}{r} \right]_{\infty}^r$$

$$W = -G M m_0 \left[\frac{1}{r} - \frac{1}{\infty} \right]$$

$$W = -\frac{G M m_0}{r}$$

$$\frac{1}{\infty} = 0$$

$$W = -\frac{G M m_0}{r}$$

$\Downarrow$

$$U = -\frac{G M m_0}{r}$$

Potential Energy $\Downarrow\Downarrow$



Gravitational Potential

(V)

★ Ratio of potential energy per unit mass

$$V = \frac{U}{m_0}$$

$$U = -\frac{G m m_0}{r}$$

$$V = \frac{-\cancel{G m m_0}}{\cancel{r} \cancel{m_0}}$$

$$V = -\frac{G m}{r}$$

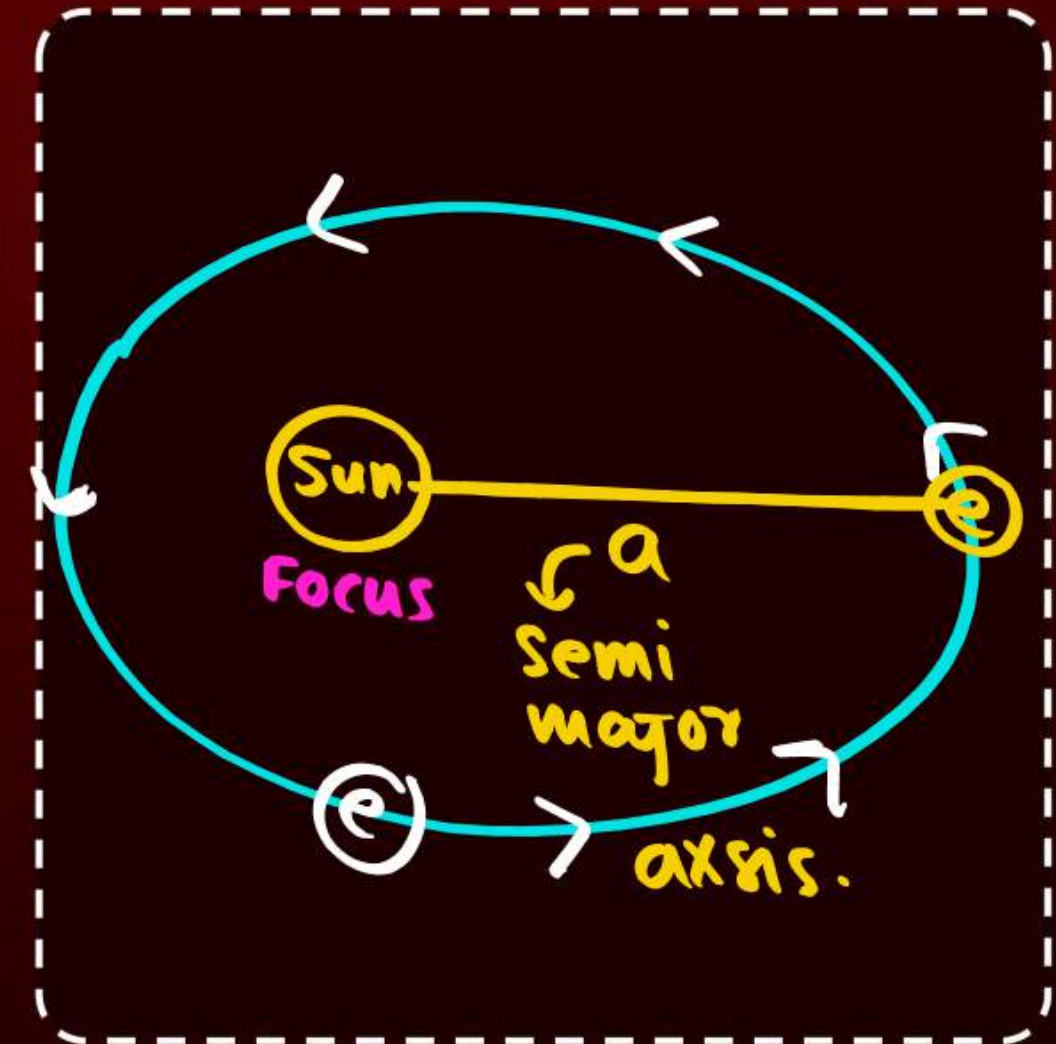
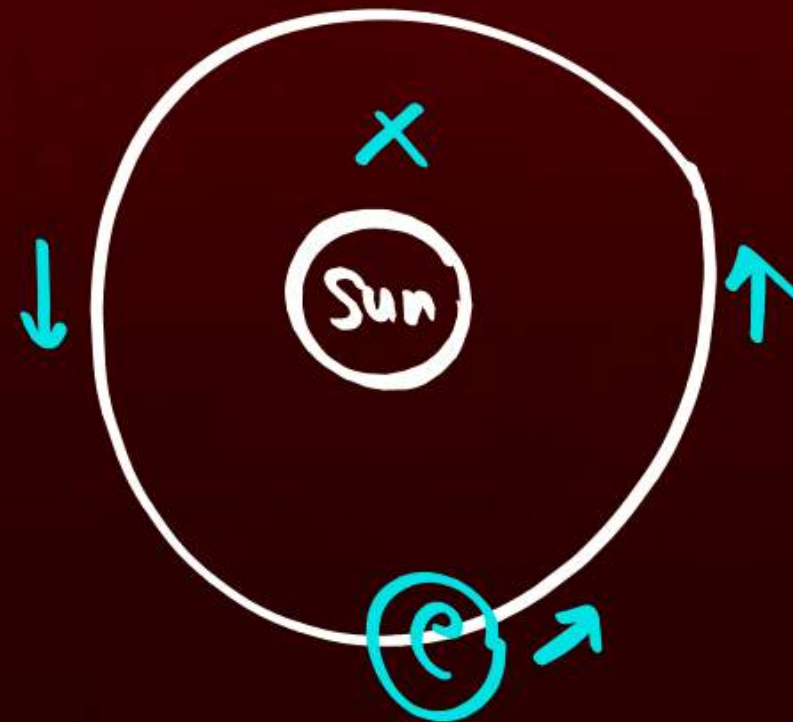




Kepler Law Of Planetary Motion

1 Law of Orbit

- It states that planet moves in elliptical orbit around Sun.





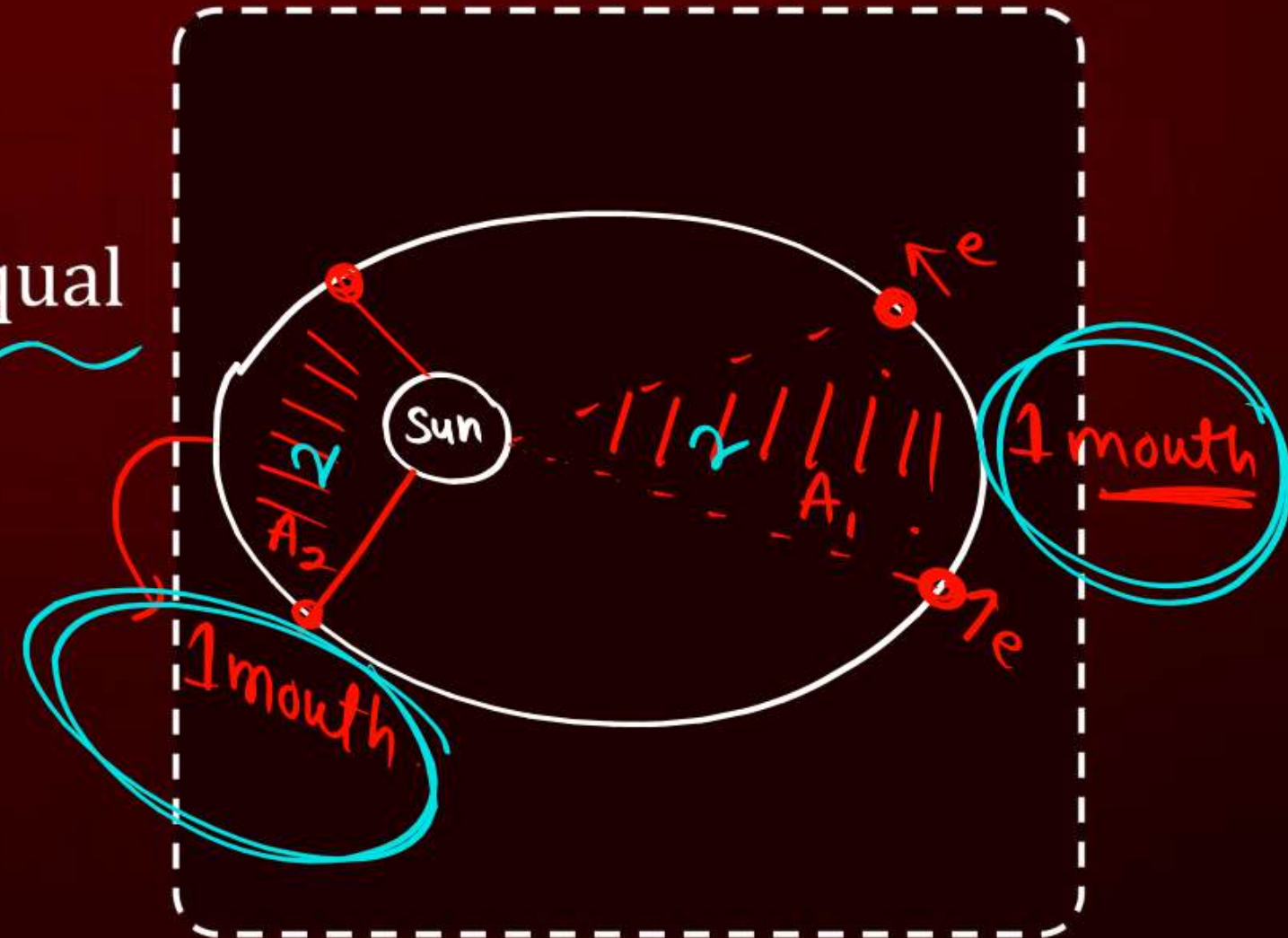
Kepler Law Of Planetary Motion

2

Law of Area

- ★ The planet sweep out equal area in equal interval of time

$$A_1 = A_2$$

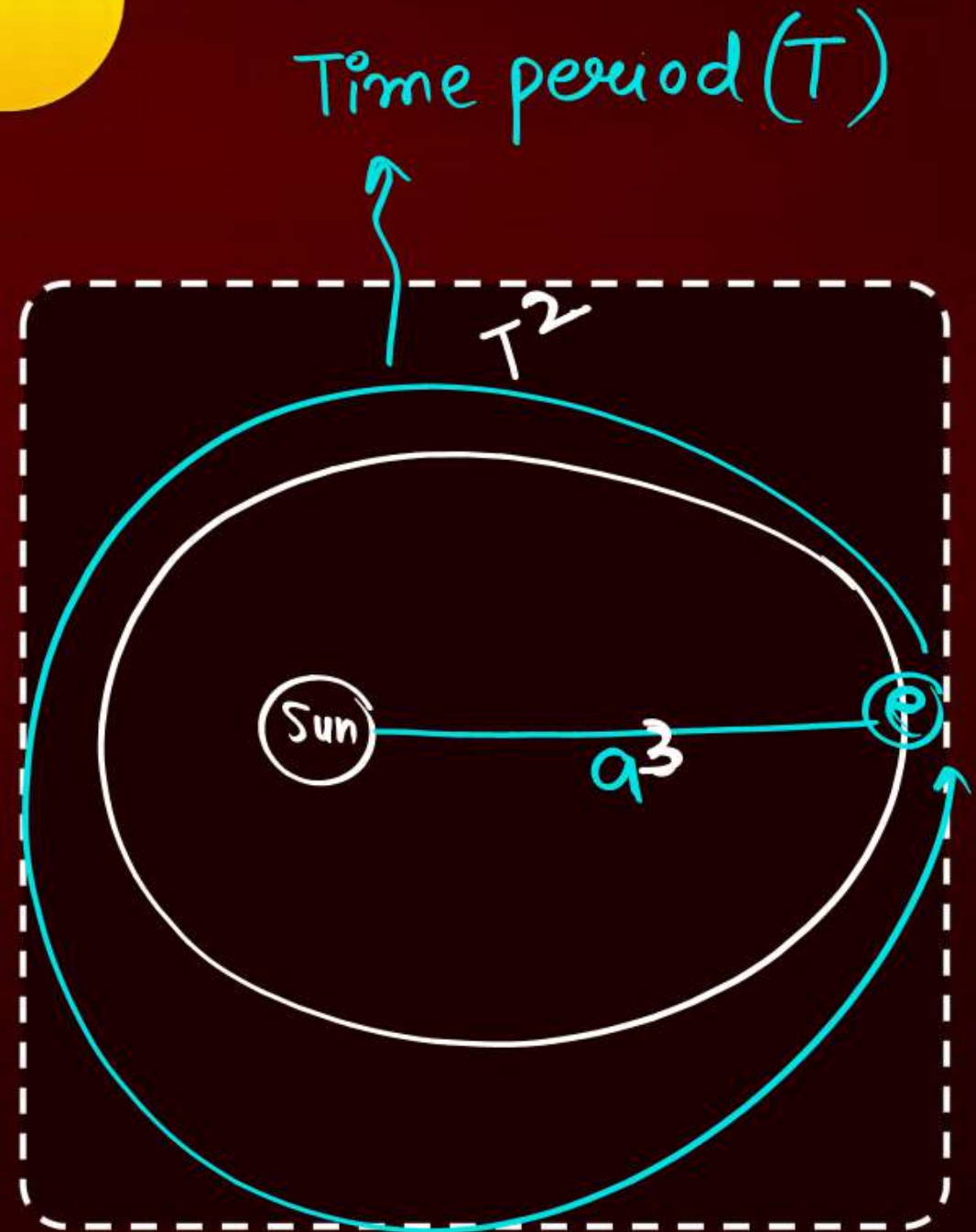




Kepler Law Of Planetary Motion

- ★ Law of Time Period
- ★ The square of time (T) is directly proportional to cube of semi major axis.

$$T^2 \propto a^3$$

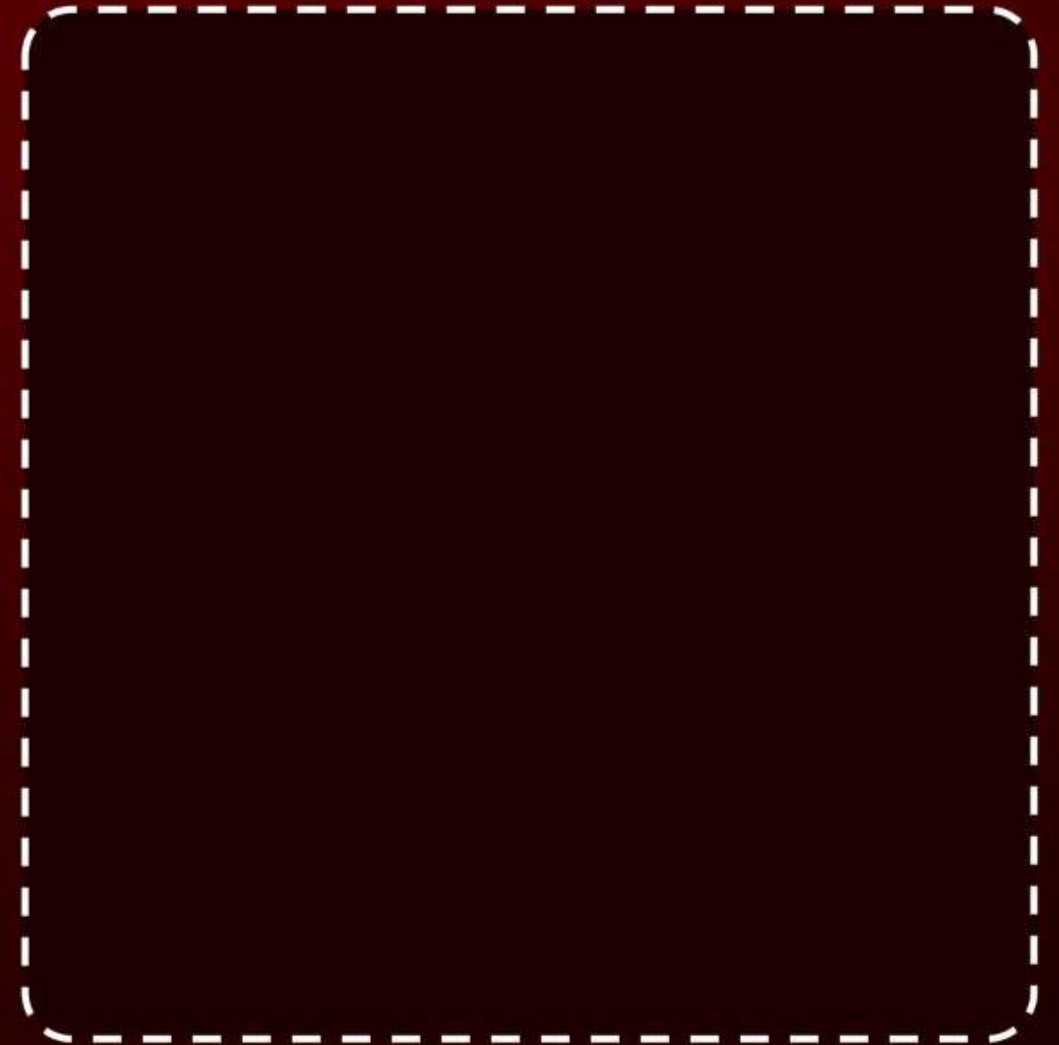




Escape Velocity

- ✦ The velocity with which you can project a object so that it leaves the Earth's gravitational field. (never come to earth again)

$$V_e = \sqrt{\frac{2Gm}{r}}$$



Derivation

Let us consider the sphere of mass (m) and radius (R) on surface (m_0) have two initial energy

$$K.E_i = \frac{1}{2} m_0 v_e^2$$

$$U_i = - \frac{G M m_0}{R}$$

In space the

$$K.E_f = 0$$

$$U_f = 0$$

Law of Conservation of Energy

$$T.E_i = T.E_f$$

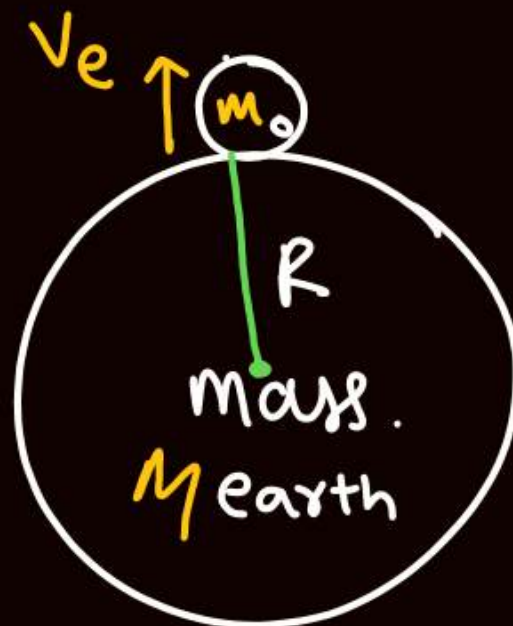
$$K.E_i + U_i = K.E_f + U_f$$

$$\frac{1}{2} m_0 v_e^2 + - \frac{G M m_0}{R} = 0 + 0$$

$$\frac{1}{2} m_0 v_e^2 = \frac{G M m_0}{R}$$

$$\frac{v_e^2}{2} = \frac{G M}{R}$$

Space $\rightarrow V=0$ $K.E_f=0$
 $U_f=0$



$$\frac{V_e^2}{2} = \frac{GM}{R}$$

$$V_e^2 = \frac{2GM}{R}$$

$$V_e = \sqrt{\frac{2GM}{R}} \quad \text{①}$$

$$V_e = \sqrt{2gR}$$

In terms of (g)

$$* g = \frac{GM}{R^2}$$

$$g = \frac{GM}{R \times R}$$

$$gR = \frac{GM}{R}$$



Escape Velocity

$$V_e = \sqrt{\frac{2GM}{R}}$$

$$V_e = \sqrt{2gR} \quad R = 6400 \text{ Km}$$

$$V_e = \sqrt{2 \times 10 \times 6400 \text{ Km}}$$

$$V_e = 11.2 \text{ Km/s (Earth)}$$

$$\underline{V_e = 11.2 \text{ km/s}}$$



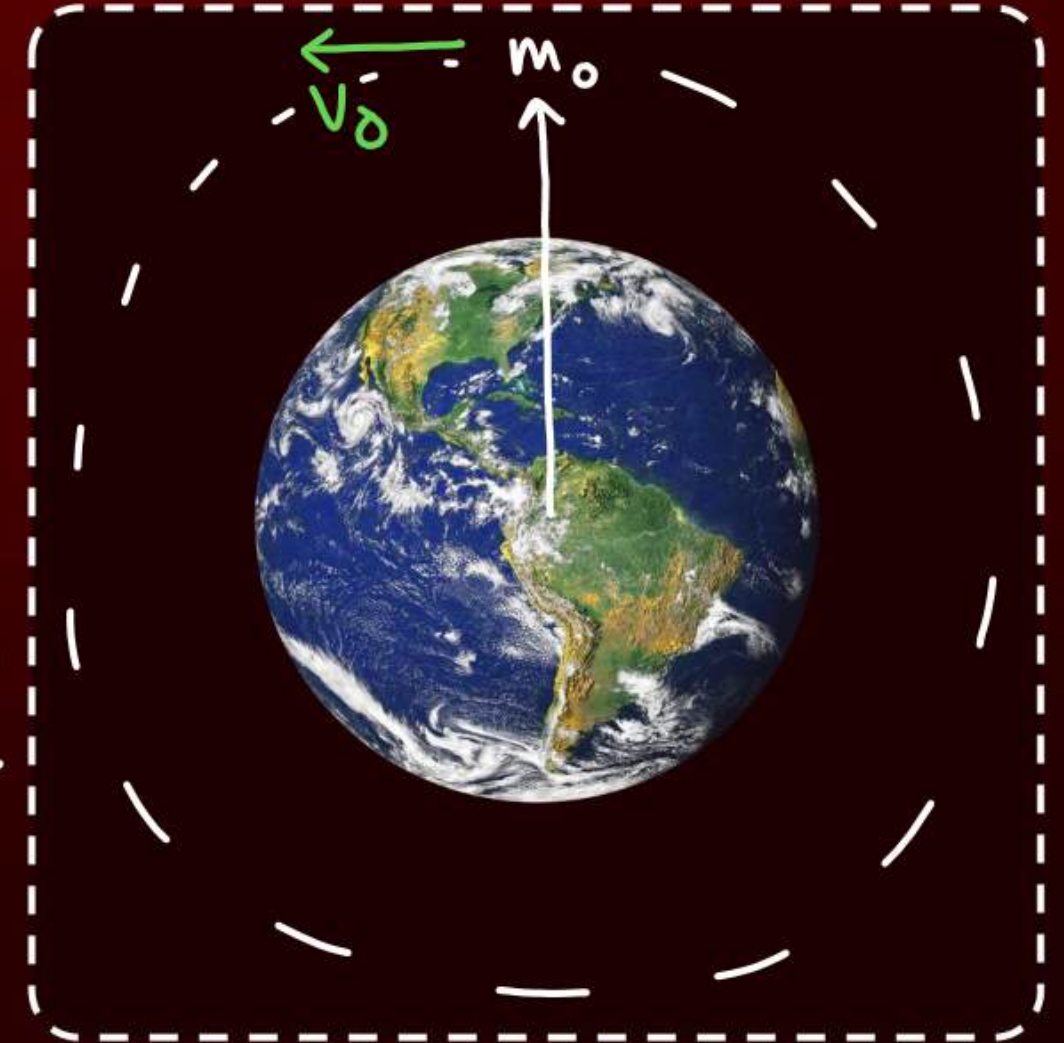


Orbital Velocity

- ✦ The heavy body revolving around another heavily body. with a velocity known as orbital velocity
- ✦ The velocity with which the satellite revolve around the planet. is called orbital velocity

$$V_0 = \sqrt{\frac{GM}{R}}$$

$$V_0 = \sqrt{gR}$$

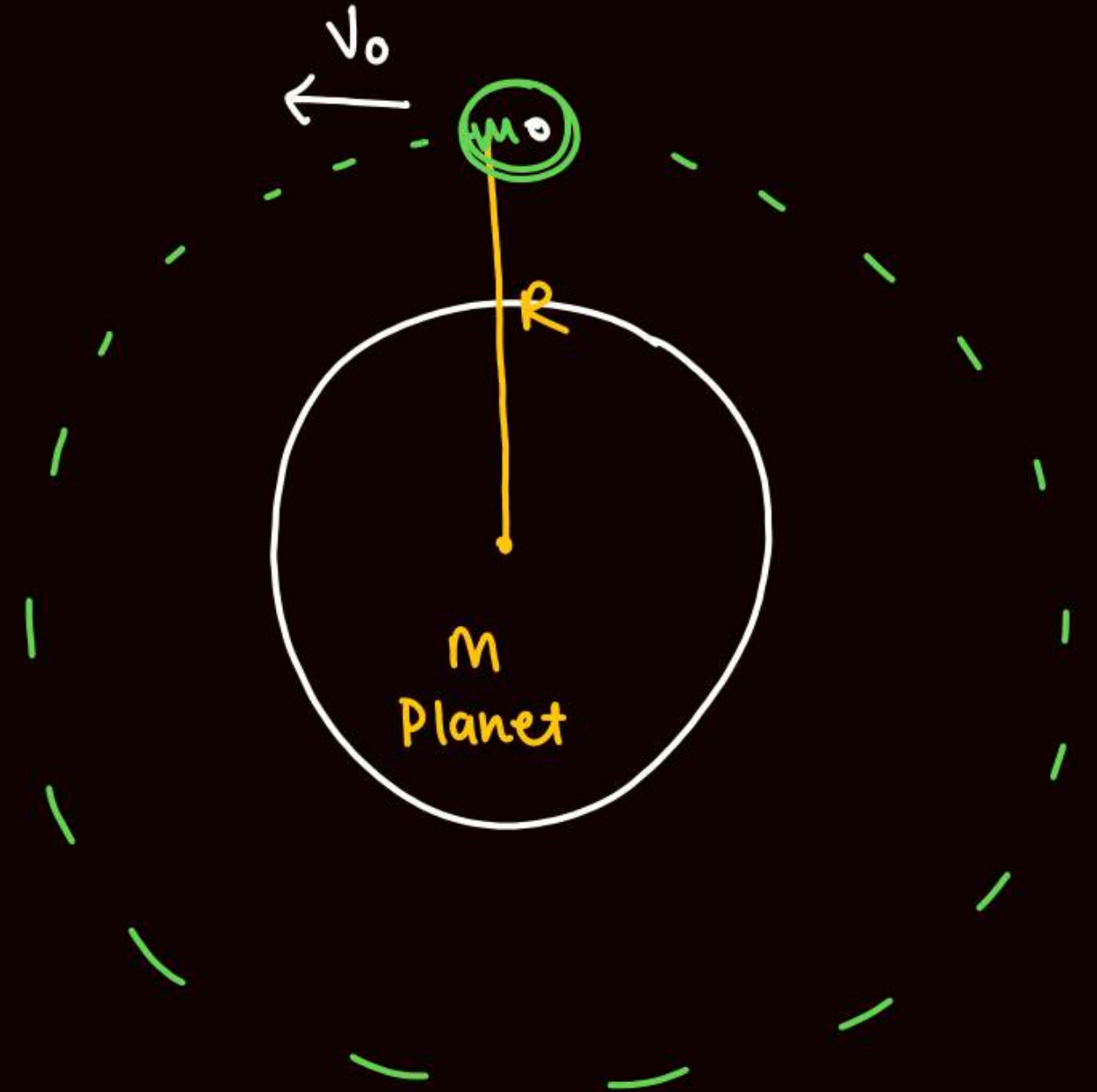


• Derivation

let us consider the planet of mass (m) and radius (R) the satellite of mass (m_0) is revolving around the planet.

The necessary centripetal force is provided by gravitation force

$$F_c = F_g$$
$$\frac{m_0 v_0^2}{R} = \frac{G M m_0}{R^2}$$

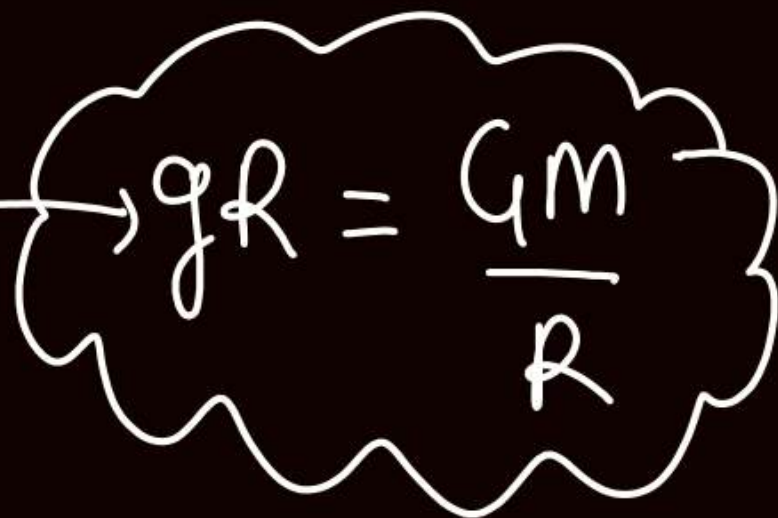


$$\frac{\cancel{m_0} v_0^2}{\cancel{R}} = \frac{G M \cancel{m_0}}{\cancel{R^2} R}$$

$$v_0^2 = \frac{Gm}{R}$$

$$v_0 = \sqrt{\frac{Gm}{R}} \quad \text{--- (I)}$$

$$v_0 = \sqrt{gR} \quad \text{--- (II)}$$


$$gR = \frac{Gm}{R}$$

Relationship b/w V_e and V_o

PYQ 2mark 2021

$$\# V_e = \sqrt{\frac{2g_m}{R}}$$

$$V_o = \sqrt{\frac{g_m}{R}}$$

$$V_e = \sqrt{2} \sqrt{\frac{g_m}{R}}^{V_o}$$

$$V_e = \sqrt{2} V_o$$





Question [2025]



Which law describes the orbits of planets around the sun?

A Newton's law

B Kirchhoff's law

C Faraday's law

D Kepler's law ✓





Question [2025]

Define orbital velocity. Derive an expression for it.





Question [2024]

Define escape velocity. Derive an expression for it.



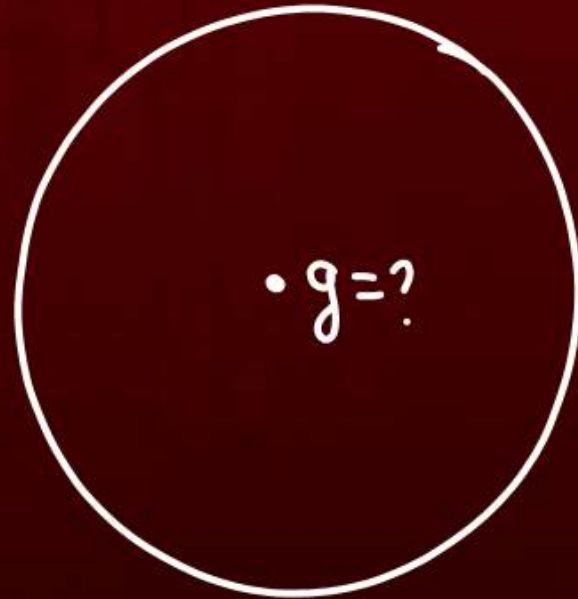


Question

H/W



What is the value of g ? At the centre of earth



$$g = \frac{Gm}{R^2} = 0$$

$$g = \frac{G \times 0}{R^2}$$

$$g = 0 \text{ m/s}^2$$

2023

2016

2018

2019



**Thank
You**